

Reinforcement Learning and Graph Neural Networks for Graph-Based Combinatorial Optimization

A Case Study of the Minimum Set Cover Problem

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Combinatorial Optimization [1]

Combinatorial Optimization

Combinatorial optimization selects a solution from a large, discrete solution space to optimize an objective function.

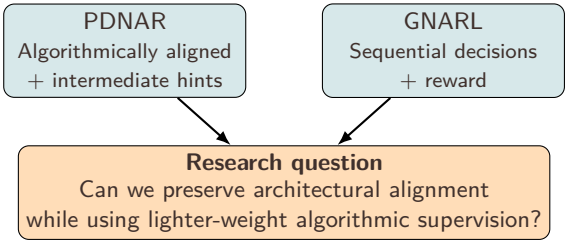
Common approaches to CO problems:

- Exact methods
- Heuristics and approximation algorithms
- Metaheuristics
- Machine-learning-based methods
  - Imitate known algorithms
  - Improve known algorithms
  - Discover new algorithms



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Research Motivation: Two Sources of Inductive Bias



Hypothesis

For an NP-hard problem, if the reference algorithm is only an approximation, forcing the network to reproduce all of its intermediate states may introduce **an overly strong inductive bias** . Decision-level supervision may leave the model more freedom to discover effective policies.

Main claim: We preserve architectural alignment without requiring the model to reproduce the algorithmic trajectory.



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The Proposed Approach

Algorithmic Alignment + Decision-Level Supervision + RL Exploration

$\Downarrow$

GNARL-MSC



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Minimum Set Cover Problem (MSC)

Problem definition

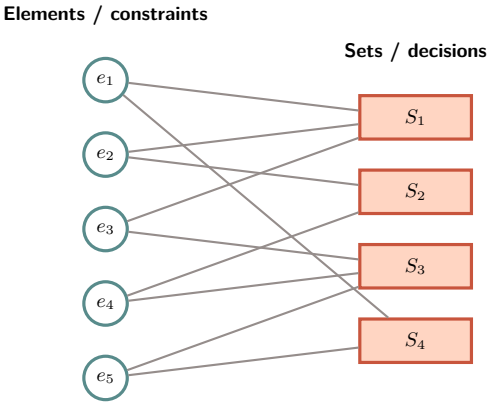
Let the universe be $U = \{e_1, \ldots, e_n\}$ and let the family of sets be $\mathcal{C} = \{S_1, \ldots, S_m\}$. The goal is to find a subfamily $A \subseteq \mathcal{C}$ of minimum total cost such that every element of U is covered:

$$\min_{A \subseteq \mathcal{C}} \sum_{S_j \in A} w_j \qquad \text{s.t.} \qquad \bigcup_{S_j \in A} S_j = U.$$



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Bipartite Graph Representation of MSC



Primal–Dual Algorithm for MSC

Algorithm 1 Batch primal–dual approximation for weighted MSC [2]

Require: Universe U, set family $\mathcal{C}$, costs $(w_S)_{S\in\mathcal{C}}$
Ensure: A set cover A

$A \leftarrow \varnothing$; $r_S \leftarrow w_S$ for all $S \in \mathcal{C}$
while $V := \{e \in U : e \notin \bigcup_{S\in A} S\} \neq \varnothing$ **do**
  repeat
    for all $e \in V$ **do**
      $\delta_e \leftarrow \min_{S \ni e} \dfrac{r_S}{|S \cap V|}$
    end for
    for all $S \in \mathcal{C} \setminus A$ **do**
      $r_S \leftarrow r_S - \sum_{e\in S\cap V} \delta_e$
    end for
  until there exists $S \in \mathcal{C} \setminus A$ with $r_S = 0$
  $A \leftarrow A \cup \{S \in \mathcal{C} : r_S = 0\}$
end while
Return A



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Linear Programming: Primal and Dual

Primal

$$\min_{x \geq 0} \quad \sum_{j=1}^{m} w_j x_j$$

$$\text{s.t.} \quad \sum_{j : e_i \in S_j} x_j \geq 1 \quad \forall e_i \in U.$$

$x_j \in \{0, 1\}$ in the primal problem

Dual

$$\max_{y \geq 0} \quad \sum_{i=1}^{n} y_i$$

$$\text{s.t.} \quad \sum_{i : e_i \in S_j} y_i \leq w_j \quad \forall S_j \in \mathcal{C}.$$

y_i : price of element e_i



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Key Step: Computing δ for elements

Current state

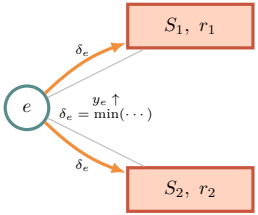
Let V be the set of uncovered elements. For each $e \in V$, the increase in its price must be small enough that no dual constraint is violated:

$$\boxed{\delta_e = \min_{S_j \ni e} \frac{r_j}{|S_j \cap V|}}$$

Then:

$$y_e \leftarrow y_e + \delta_e.$$

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Simultaneous updates

In the batched version, δ_e is computed for every element in V ; then all price increases are applied to the residual costs of the sets.



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Relaxing Residual Costs to Zero

Updating Set Residuals

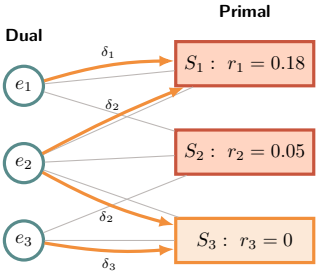
After increasing the dual prices, for each unselected set:

$$r_j \leftarrow r_j - \sum_{e \in S_j \cap V} \delta_e$$

This process continues until at least one set reaches $r_j = 0$.



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NAR

Two different questions

- In standard learning, a model directly learns a mapping from inputs to outputs:

$$x \longmapsto y.$$

- In Neural Algorithmic Reasoning, the model attempts to execute the **computational process** of an algorithm:

$$s^{(0)} \longrightarrow s^{(1)} \longrightarrow \cdots \longrightarrow s^{(T)}.$$

Core idea of NAR

Rather than learning only an input-output correlation, the neural network imitates the step-by-step computation of the algorithm in its latent representations.



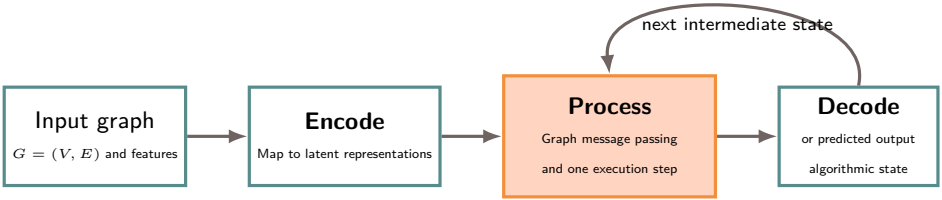
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NAR (continued)

Objectives

- Learn to execute an algorithm without reducing the data dimension
- Reconstruct intermediate computational states
- Achieve out-of-distribution generalization
- Transfer learned embeddings to other tasks

The Encode-Process-Decode Pattern



Interpretation

Node, edge, and graph features are first mapped to a latent space. A graph processor then performs computation by passing messages across neighborhoods and reconstructs the output of the current step.



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Supervision with hints

Operational definition

Within NAR, a hint is an algorithmic feature or internal state that changes during execution and is used to supervise the network at intermediate steps.

$$\widehat{y}^{(t)} = \mathrm{Decode}\Big(h^{(t)}\Big) \approx y^{(t)}_{\text{algorithm}}$$

Training effect

The network is not evaluated only against the final answer; it must also reconstruct the algorithm's intermediate states.

Stronger alignment $\iff$ Less freedom to discover solutions



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An Intuitive Definition of Algorithmic Alignment

Definition

An architecture is **algorithmically aligned** with an algorithm if its computation pattern, connectivity, and updates are consistent with the algorithm's computational structure .

Expected outcome

The more closely the architecture matches the target computation, the fewer examples the network may need to learn algorithmic behavior, potentially improving size generalization .



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Application Example: Dual Algorithmic Reasoning [3]

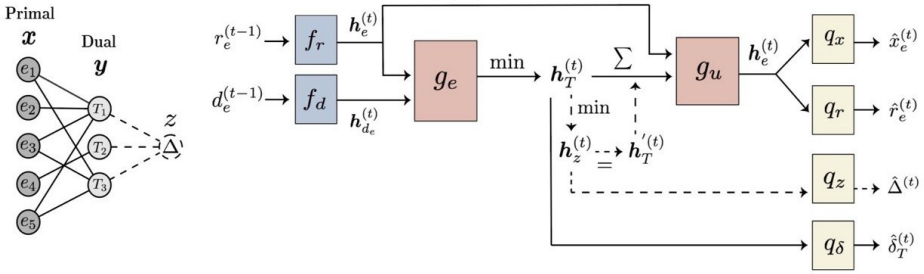
Idea

In the maximum-flow problem, Max-Flow and Min-Cut are are two related views of the same structure. A neural model can learn both computations and use information from one to improve the other.

Real-world example

In the study Dual Algorithmic Reasoning [3]
the combination of learning max-flow and min-cut was studied for cerebral vessel classification; flow structure can provide information unavailable from direct labeling alone.

Primal-Dual Algorithmic Reasoning (PDNAR)



Conceptual interpretation

PDNAR does not only say "which sets to select"; it also partly specifies "which dual state and residual costs should lead to the selection."



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Results and Limitations of PDNAR for MSC

Strength

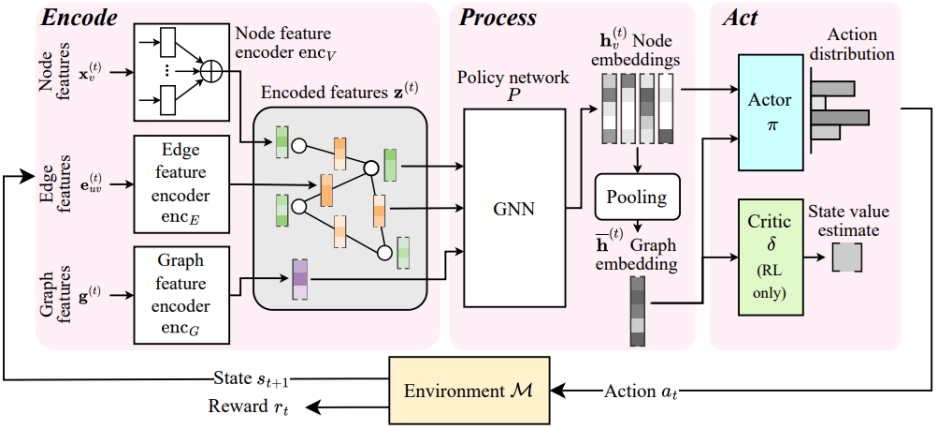
Combining a bipartite graph, primal–dual information, and stepwise supervision creates a strong algorithmic inductive bias for set cover.

Research question

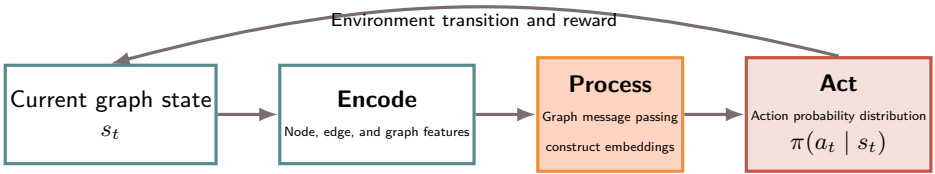
If the algorithmic trajectory is only approximate, does fully supervising that trajectory overly constrain the model?

Can the model discover a better solution to the problem instead of imitating the approximation?

GNARL: Reformulating NAR as MDP [4]



The Encode-Process-Act Pattern





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Action Space and Solution Validity in GNARL

Decision model

$$\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, h \rangle$$

Action selection

- The network produces a score for each node.
- Scores are converted into a probability distribution.
- Sampling or greedy selection can be used.

Action masking

$$\mathcal{A}(s) \subseteq \mathcal{V}$$

- Invalid actions are masked out.
- For example, a selected set cannot be selected again.
- The solution is constructed to be valid from the outset.

Difference from post-processing

In this design, solution validity does not depend on extensive search or a separate cleanup stage; the environment permits only valid actions.



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Training GNARL: Imitation and Reinforcement Learning

Behavioral Cloning

If an expert trajectory is available:

$$\mathcal{D} = \{(s_t, a_t^{\text{expert}})\}$$

the policy is trained by minimizing classification loss:

$$\mathcal{L}_{\text{BC}} = -\mathbb{E}\left[\log \pi_\theta(a_t^{\text{expert}} \mid s_t)\right].$$

Reinforcement Learning

Without trajectory labels, the model learns from environmental rewards:

$$\max_\theta \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{h-1} \gamma^t r_t\right].$$

GNARL uses methods such as PPO.

Combining two signals

Expert imitation can provide a useful initialization; reinforcement learning can then improve the policy based on actual solution quality.



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Examples of Reported Results for GNARL

Tasks with Expert Algorithms

For algorithmic tasks such as CLRS, training by imitating expert trajectories can achieve high accuracy; GNARL can also generate multiple valid solutions by sampling its policy.

Combinatorial optimization problems

The framework has been evaluated on problems such as minimum vertex cover and the traveling salesperson problem.

In the GNARL paper, both imitation-trained models and models trained with PPO achieved reasonable performance on instances larger than those used for training .

Main message

GNARL aims to provide a common framework for polynomial-time and NP-hard problems without necessarily depending on a specific approximation algorithm .



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Section Summary: Research Idea

Graph structure of the problem

$+$ Alignment with primal–dual computation

$+$ Decision-level supervision

$\Downarrow$

Architecture GNARL for MSC

Perspective

We aim to preserve the useful inductive bias of an algorithm-aligned architecture without forcing the network to reconstruct every intermediate state of an approximation algorithm.

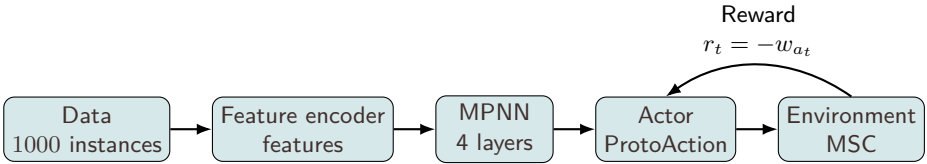
Empirical question

Can replacing supervision on hints with decision-level supervision improve solution quality and the discovery of effective solutions while preserving a substantial part of size generalization?



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From Idea to Implementation





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Decision Formulation: One Selection per Step

State

The state includes the bipartite graph, selected sets, covered elements, and latent node representations.

Valid action

$$\mathcal{A}(s_t) = \left\{ S_j \mid x_j = 0,\ S_j \cap U_t^{\text{uncovered}} \neq \varnothing \right\}$$

Thus, a previously selected set or a set that covers no new element cannot be selected.

Key consequence

Solution validity is largely ensured during the **construction process** . The model need not first produce an arbitrary vector and then repair it.



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Architecture GNARL-MSC

Graph processor

- Element-set bipartite graph
- Bidirectional message passing
- Update representations with GRU
- 4 message-passing layers
- Hidden dimension: 64
- mean pooling for graph-level representation

Decision module

- Actor: ProtoAction
- Action score based on the distance between the set embedding and the action-prototype vector
- Dynamic masking of invalid actions
- Critic: MLP over the graph-level representation



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Architectural Alignment with Primal-Dual Computation

What the architecture actually supports

- The bipartite graph preserves the structure of the coverage constraints.
- Sum aggregation enables summation over adjacent elements or sets.
- Bidirectional message passing matches local dependencies in primal-dual computation.
- The network has the **representational capacity** to encode quantities related to dual computation.

What we do not claim

- The network does not necessarily execute the primal-dual algorithm.
- Primal-Dual constraints are not enforced.
- Therefore, this is **algorithmic alignment** , not proof that the network is equivalent to the approximation algorithm.



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Architectural Alignment with Primal-Dual Computation

Set → Element message

$$M_i^{(t)} = \sum_{j:\,(i,j)\in E} \phi_{SE}^{(t)}\left(h_j^{(t)}\right)$$

$$h_i^{(t+1)} = \mathrm{GRU}_E^{(t)}\left(M_i^{(t)}, h_i^{(t)}\right)$$

Element → Set message

$$N_j^{(t)} = \sum_{i:\,(i,j)\in E} \phi_{ES}^{(t)}\left(h_i^{(t+1)}\right)$$

$$h_j^{(t+1)} = \mathrm{GRU}_S^{(t)}\left(N_j^{(t)}, h_j^{(t)}\right)$$

Correspondence with primal–dual computation

For each set, a quantity of the form $\sum_{i\in S_j} y_i$ requires **summing over the elements adjacent to that set**, exactly the the operation performed along the element→set message-passing direction.



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Data and Experimental Protocol

Component	Setting
Problem	Weighted Minimum Set Cover
Representation	Element–set bipartite graph
Training data generation	1000 bipartite graph BA
Training size	16
Training parameter	$b = 5$
Sizes ID	$16, 32, 64, 128, 256, 512, 1024$
OOD sizes	$16, 128, 512, 1024$, with $b = 3, 8$
Instances per size	100
Evaluation seeds	5

Scale note

The largest test instance is $64\times$ the training size, so the evaluation is not merely interpolation.



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Training setup: BC then PPO

$$r_t = -w_{a_t} \qquad \Longrightarrow \qquad \sum_t r_t = -J(\hat{x}) \quad \text{(for a complete episode)}$$

1. **BC:** The model first learns the mapping "state → expert decision" on small instances.
2. **Limitation of BC:** At test time, one incorrect decision can move the policy into states outside the expert distribution.
3. **PPO:** The agent now observes states induced by its own decisions and interacts directly through rewards.
4. **Conceptual implication:** The expert acts as a *prior* ; the final policy need not simply copy the expert.

$$\underbrace{\text{Expert decisions}}_{\text{supervision}} \rightarrow \underbrace{\text{BC policy}}_{\text{initialization}} \rightarrow \underbrace{\text{PPO}}_{\text{search / adaptation}}$$



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Evaluation Metrics

Cost

$$J(\hat{x}) = \sum_j w_j \hat{x}_j$$

$$\rho_{\text{alg}} = \frac{J(\hat{x}_{model})}{J(x_{approx})} \qquad \rho_\star = \frac{J(\hat{x}_{model})}{J(x^\star)}$$

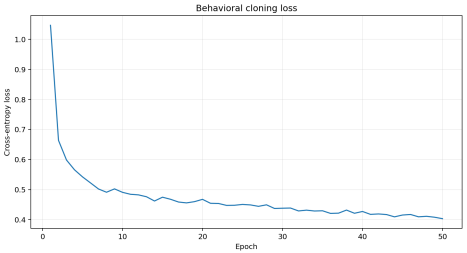
Interpretive caution

ρ_{alg} is an empirical ratio to a baseline, not a theoretical approximation guarantee.



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Training Curve: Convergence





Interpretation

PPO does not appear to discover better decisions as easily as BC . Compute constraints prevented training the model for more epochs.



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Key Result: In-Distribution Size Generalization

Method	16	32	64	128	256	512	1024
PDNAR-NoAlgo	1.028	1.025	1.010	1.017	1.012	1.008	1.006
PDNAR-NoOptm	1.008	1.008	.997	.992	.981	.973	.975
PDNAR	.979	.918	.947	.915	.920	.915	.913
GNARL-BC	.909	.825	.814	.818	.798	.789	.785
GNARL-PPO	**.908**	**.815**	**.810**	**.813**	**.794**	**.779**	**.774**

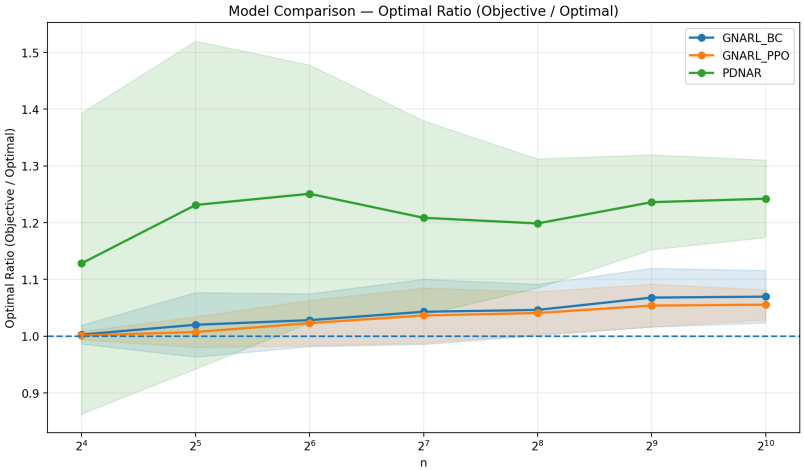
Observation

Under this protocol, GNARL-PPO has a lower cost ratio than the approximation algorithm and the reported PDNAR baseline. At size 1024, a ratio of 0.774 means about 22.6% lower cost than the approximation baseline.



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In-Distribution Visualization





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Out-of-Distribution Generalization: Structural and Size

Test

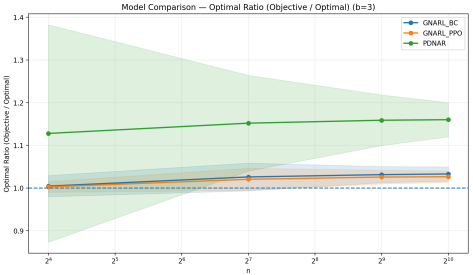
The model was trained with $b = 5$ and evaluated with $b = 3$ and $b = 8$. Changing b changes the degree distribution and the density of membership relations.

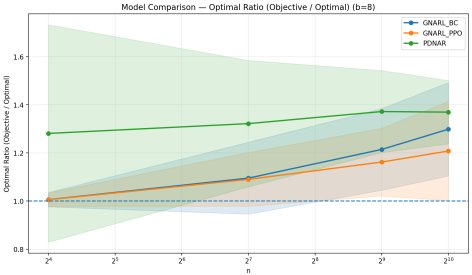
Method / distribution	16	128	512	1024
PDNAR, $b = 3$	.943	.918	.929	.922
GNARL-PPO, $b = 3$	**.866**	**.839**	**.819**	**.822**
PDNAR, $b = 8$	.969	.940	.941	.943
GNARL-PPO, $b = 8$	**.854**	**.811**	**.819**	**.841**

Result

The empirical advantage of GNARL-PPO persists under both distribution shifts; however, higher variance and the limited number of seeds constrain the strength of this claim.

Visualization OOD





The plots show the ratio $J/J^\star$; the table in previous slide reports J/J_{approx} .



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Is the Hypothesis Supported?

Supporting evidence

- Without intermediate hints, the model achieves competitive performance across sizes from 16 to 1024.
- Under the current protocol, GNARL-PPO outperforms the approximation baseline and the reported PDNAR results.
- The model also remains competitive under OOD shifts in b.

What remains unproven

- Whether replacing hints with expert behavioral cloning actually causes the improvement.
- Whether generalization holds at much larger sizes.
- Whether results are robust to random seeds, reward design, and the data generator.



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Summary

Architectural alignment
+ Decision-level supervision
+ Exploration with RL

$\Longrightarrow$ GNARL-MSC

- **Idea:** We preserve the architecture's inductive bias without requiring full reconstruction of the approximation algorithm's trajectory.
- **Implementation:** bipartite graph + MPNN + ProtoAction + action masking + BC/PPO.
- **Finding:** In the current experiments, solution quality relative to the approximation baseline and reported PDNAR reference values is promising, and size generalization is observed.
- **Scientific position:** The result is meaningful, but stronger controlled experiments are needed to make causal claims about removing hints and about general superiority.



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